

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv("netflix1.csv")
```

```
df
```

|      | show_id | type    | title                            | director        | country       | date_added | release_year | rating | duration  | listed_in                                         |  |
|------|---------|---------|----------------------------------|-----------------|---------------|------------|--------------|--------|-----------|---------------------------------------------------|--|
| 0    | s1      | Movie   | Dick Johnson Is Dead             | Kirsten Johnson | United States | 9/25/2021  | 2020         | PG-13  | 90 min    | Documentaries                                     |  |
| 1    | s3      | TV Show | Ganglands                        | Julien Leclercq | France        | 9/24/2021  | 2021         | TV-MA  | 1 Season  | Crime TV Shows, International TV Shows, TV Act... |  |
| 2    | s6      | TV Show | Midnight Mass                    | Mike Flanagan   | United States | 9/24/2021  | 2021         | TV-MA  | 1 Season  | TV Dramas, TV Horror, TV Mysteries                |  |
| 3    | s14     | Movie   | Confessions of an Invisible Girl | Bruno Garotti   | Brazil        | 9/22/2021  | 2021         | TV-PG  | 91 min    | Children & Family Movies, Comedies                |  |
| 4    | s8      | Movie   | Sankofa                          | Haile Gerima    | United States | 9/24/2021  | 1993         | TV-MA  | 125 min   | Dramas, Independent Movies, International Movies  |  |
| ...  | ...     | ...     | ...                              | ...             | ...           | ...        | ...          | ...    | ...       | ...                                               |  |
| 8785 | s8797   | TV Show | Yunus Emre                       | Not Given       | Turkey        | 1/17/2017  | 2016         | TV-PG  | 2 Seasons | International TV Shows, TV Dramas                 |  |
| 8786 | s8798   | TV Show | Zak Storm                        | Not Given       | United States | 9/13/2018  | 2016         | TV-Y7  | 3 Seasons | Kids' TV                                          |  |
| 8787 | s8801   | TV Show | Zindagi Gulzar Hai               | Not Given       | Pakistan      | 12/15/2016 | 2012         | TV-PG  | 1 Season  | International TV Shows, Romantic TV Shows, TV ... |  |
| 8788 | s8784   | TV Show | Yoko                             | Not Given       | Pakistan      | 6/23/2018  | 2016         | TV-Y   | 1 Season  | Kids' TV                                          |  |
| 8789 | s8786   | TV      | YOM                              | Not Given       | Pakistan      | 6/27/2018  | 2016         | TV-Y7  | 1 Season  | Kids' TV                                          |  |

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.head()
```

|     | show_id | type    | title                            | director        | country       | date_added | release_year | rating | duration | listed_in                                         |  |
|-----|---------|---------|----------------------------------|-----------------|---------------|------------|--------------|--------|----------|---------------------------------------------------|--|
| 0   | s1      | Movie   | Dick Johnson Is Dead             | Kirsten Johnson | United States | 9/25/2021  | 2020         | PG-13  | 90 min   | Documentaries                                     |  |
| 1   | s3      | TV Show | Ganglands                        | Julien Leclercq | France        | 9/24/2021  | 2021         | TV-MA  | 1 Season | Crime TV Shows, International TV Shows, TV Act... |  |
| 2   | s6      | TV Show | Midnight Mass                    | Mike Flanagan   | United States | 9/24/2021  | 2021         | TV-MA  | 1 Season | TV Dramas, TV Horror, TV Mysteries                |  |
| 3   | s14     | Movie   | Confessions of an Invisible Girl | Bruno Garotti   | Brazil        | 9/22/2021  | 2021         | TV-PG  | 91 min   | Children & Family Movies, Comedies                |  |
| ... | ...     | ...     | ...                              | ...             | ...           | ...        | ...          | ...    | ...      | ...                                               |  |

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail()
```

|      | show_id | type    | title              | director  | country       | date_added | release_year | rating | duration  | listed_in                                         |  |
|------|---------|---------|--------------------|-----------|---------------|------------|--------------|--------|-----------|---------------------------------------------------|--|
| 8785 | s8797   | TV Show | Yunus Emre         | Not Given | Turkey        | 1/17/2017  | 2016         | TV-PG  | 2 Seasons | International TV Shows, TV Dramas                 |  |
| 8786 | s8798   | TV Show | Zak Storm          | Not Given | United States | 9/13/2018  | 2016         | TV-Y7  | 3 Seasons | Kids' TV                                          |  |
| 8787 | s8801   | TV Show | Zindagi Gulzar Hai | Not Given | Pakistan      | 12/15/2016 | 2012         | TV-PG  | 1 Season  | International TV Shows, Romantic TV Shows, TV ... |  |

```
df.sample()
```

|      | show_id | type  | title                      | director           | country | date_added | release_year | rating | duration | listed_in                              |
|------|---------|-------|----------------------------|--------------------|---------|------------|--------------|--------|----------|----------------------------------------|
| 2917 | s3951   | Movie | Ek Ladki Ko Dekha Toh Aisa | Shelly Chopra Dhar | India   | 4/2/2019   | 2019         | TV-14  | 120 min  | Comedies, Dramas, International Movies |

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   show_id               8790 non-null   object
1   type                  8790 non-null   object
2   title                 8790 non-null   object
3   director              8790 non-null   object
4   country               8790 non-null   object
5   date_added            8790 non-null   object
6   release_year          8790 non-null   int64
7   rating                8790 non-null   object
8   duration              8790 non-null   object
9   listed_in             8790 non-null   object
dtypes: int64(1), object(9)
memory usage: 686.8+ KB
```

```
df.describe()
```

|       | release_year |
|-------|--------------|
| count | 8790.000000  |
| mean  | 2014.183163  |
| std   | 8.825466     |
| min   | 1925.000000  |
| 25%   | 2013.000000  |
| 50%   | 2017.000000  |
| 75%   | 2019.000000  |
| max   | 2021.000000  |

```
df.index
```

```
RangeIndex(start=0, stop=8790, step=1)
```

```
df.shape
```

```
(8790, 10)
```

```
df.size
```

```
87900
```

```
df.isnull().sum()
```

|              | 0 |
|--------------|---|
| show_id      | 0 |
| type         | 0 |
| title        | 0 |
| director     | 0 |
| country      | 0 |
| date_added   | 0 |
| release_year | 0 |
| rating       | 0 |
| duration     | 0 |
| listed_in    | 0 |

```
dtype: int64
```

```
df.rating.unique()
```

```
array(['PG-13', 'TV-MA', 'TV-PG', 'TV-14', 'TV-Y7', 'TV-Y', 'PG', 'TV-G',
      'R', 'G', 'NC-17', 'NR', 'TV-Y7-FV', 'UR'], dtype=object)
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['show_id'] = df['show_id'].str.replace('s', '')
df.head()
```

|   | show_id | type    | title                            | director        | country       | date_added | release_year | rating | duration | listed_in                                         |  |
|---|---------|---------|----------------------------------|-----------------|---------------|------------|--------------|--------|----------|---------------------------------------------------|--|
| 0 | 1       | Movie   | Dick Johnson Is Dead             | Kirsten Johnson | United States | 9/25/2021  | 2020         | PG-13  | 90 min   | Documentaries                                     |  |
| 1 | 3       | TV Show | Ganglands                        | Julien Leclercq | France        | 9/24/2021  | 2021         | TV-MA  | 1 Season | Crime TV Shows, International TV Shows, TV Act... |  |
| 2 | 6       | TV Show | Midnight Mass                    | Mike Flanagan   | United States | 9/24/2021  | 2021         | TV-MA  | 1 Season | TV Dramas, TV Horror, TV Mysteries                |  |
| 3 | 14      | Movie   | Confessions of an Invisible Girl | Bruno Garotti   | Brazil        | 9/22/2021  | 2021         | TV-PG  | 91 min   | Children & Family Movies, Comedies                |  |
| 4 | 8       | Movie   | Sankofa                          | Haile Gerima    | United States | 9/24/2021  | 1993         | TV-MA  | 125 min  | Dramas, Independent Movies, International         |  |

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.groupby('country')['country'].count().sort_values(ascending=False).head(5)
```

| country        |      |
|----------------|------|
| country        |      |
| United States  | 3240 |
| India          | 1057 |
| United Kingdom | 638  |
| Pakistan       | 421  |
| Not Given      | 287  |

dtype: int64

```
df.groupby('country')['country'].count().sort_values(ascending=False).tail(5)
```

| country      |   |
|--------------|---|
| country      |   |
| Slovenia     | 1 |
| Puerto Rico  | 1 |
| Somalia      | 1 |
| West Germany | 1 |
| Zimbabwe     | 1 |

dtype: int64

```
df['type'].value_counts()
```

| count   |      |
|---------|------|
| type    |      |
| Movie   | 6126 |
| TV Show | 2664 |

dtype: int64

```
df.loc[df['release_year'] == 2020]
```

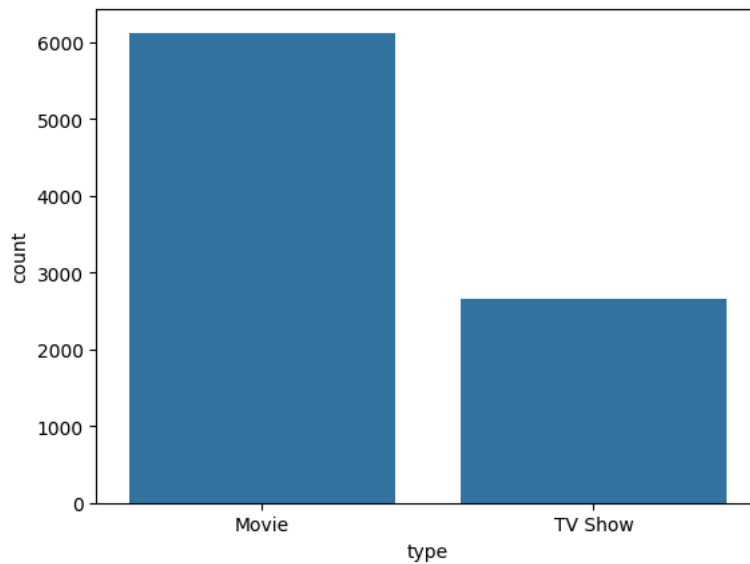
|      | show_id | type    | title                                             | director                                      | country       | date_added | release_year | rating | duration  | listed_in                                         |
|------|---------|---------|---------------------------------------------------|-----------------------------------------------|---------------|------------|--------------|--------|-----------|---------------------------------------------------|
| 0    | 1       | Movie   | Dick Johnson Is Dead                              | Kirsten Johnson                               | United States | 9/25/2021  | 2020         | PG-13  | 90 min    | Documentaries                                     |
| 25   | 17      | Movie   | Europe's Most Dangerous Man: Otto Skorzeny in ... | Pedro de Echave García, Pablo Azorín Williams | Not Given     | 9/22/2021  | 2020         | TV-MA  | 67 min    | Documentaries, International Movies               |
| 36   | 35      | TV Show | Tayo and Little Wizards                           | Not Given                                     | Pakistan      | 9/17/2021  | 2020         | TV-Y7  | 1 Season  | Kids' TV                                          |
| 47   | 190     | TV Show | Bread Barbershop                                  | Not Given                                     | Pakistan      | 8/28/2021  | 2020         | TV-Y   | 2 Seasons | Kids' TV, TV Comedies                             |
| 58   | 48      | TV Show | The Smart Money Woman                             | Bunmi Ajakaiye                                | South Africa  | 9/16/2021  | 2020         | TV-MA  | 1 Season  | International TV Shows, Romantic TV Shows, TV ... |
| ...  | ...     | ...     | ...                                               | ...                                           | ...           | ...        | ...          | ...    | ...       | ...                                               |
| 7484 | 3288    | TV Show | Maradona in Mexico                                | Not Given                                     | Argentina     | 11/13/2019 | 2020         | TV-MA  | 1 Season  | Docuseries, Spanish-Language TV Shows             |
| 7517 | 3370    | TV Show | BoJack Horseman                                   | Not Given                                     | United States | 10/25/2019 | 2020         | TV-MA  | 6 Seasons | TV Comedies                                       |
| 7537 | 3434    | TV      | The Hook Up                                       | Not Given                                     | France        | 10/11/2019 | 2020         | TV-MA  | 2         | International TV Shows, Romantic                  |

df.iloc[100:110]

|      | show_id | type    | title                                    | director              | country       | date_added | release_year | rating | duration | listed_in                                         |
|------|---------|---------|------------------------------------------|-----------------------|---------------|------------|--------------|--------|----------|---------------------------------------------------|
| 8685 | 8126    | TV      | Super Wings                              | Not Given             | United States | 12/1/2020  | 2020         | TV-Y   | 3        | Kids' TV, Korean TV Shows                         |
| 100  | 370     | TV Show | Myth & Mogul: John DeLorean              | Not Given             | Pakistan      | 7/30/2021  | 2021         | TV-14  | 1 Season | British TV Shows, Crime TV Shows, Docuseries      |
| 101  | 377     | TV Show | Transformers: War for Cybertron: Kingdom | Not Given             | Pakistan      | 7/29/2021  | 2021         | TV-Y7  | 1 Season | Anime Series                                      |
| 102  | 380     | TV Show | Tattoo Redo                              | Not Given             | Pakistan      | 7/28/2021  | 2021         | TV-MA  | 1 Season | Reality TV                                        |
| 103  | 382     | TV Show | The Snitch Cartel: Origins               | Not Given             | Pakistan      | 7/28/2021  | 2021         | TV-MA  | 1 Season | Crime TV Shows, International TV Shows, Spanis... |
| 104  | 398     | TV Show | Feels Like Ishq                          | Not Given             | Pakistan      | 7/23/2021  | 2021         | TV-MA  | 1 Season | International TV Shows, Romantic TV Shows, TV ... |
| 105  | 483     | TV Show | How to Become a Tyrant                   | Not Given             | Pakistan      | 7/9/2021   | 2021         | TV-MA  | 1 Season | Docuseries                                        |
| 106  | 82      | Movie   | Kate                                     | Cedric Nicolas-Troyan | United States | 9/10/2021  | 2021         | R      | 106 min  | Action & Adventure                                |
| 107  | 85      | Movie   | Omo Ghetto: the Saga                     | JJC Skillz, Funke     | Nigeria       | 9/10/2021  | 2020         | TV-MA  | 147 min  | Action & Adventure, Comedies                      |

## Plotting

```
sns.countplot(x='type',data=df)
plt.show()
```



```
df['country'].value_counts()
```

| count          |      |
|----------------|------|
| country        |      |
| United States  | 3240 |
| India          | 1057 |
| United Kingdom | 638  |
| Pakistan       | 421  |
| Not Given      | 287  |
| ...            | ...  |
| Luxembourg     | 1    |
| Senegal        | 1    |
| Belarus        | 1    |
| Puerto Rico    | 1    |
| Cyprus         | 1    |

86 rows × 1 columns

dtype: int64

```
top_10 = df['country'].value_counts().nlargest(10)
top_10
```

| count          |      |
|----------------|------|
| country        |      |
| United States  | 3240 |
| India          | 1057 |
| United Kingdom | 638  |
| Pakistan       | 421  |
| Not Given      | 287  |
| Canada         | 271  |
| Japan          | 259  |
| South Korea    | 214  |
| France         | 213  |
| Spain          | 182  |

dtype: int64

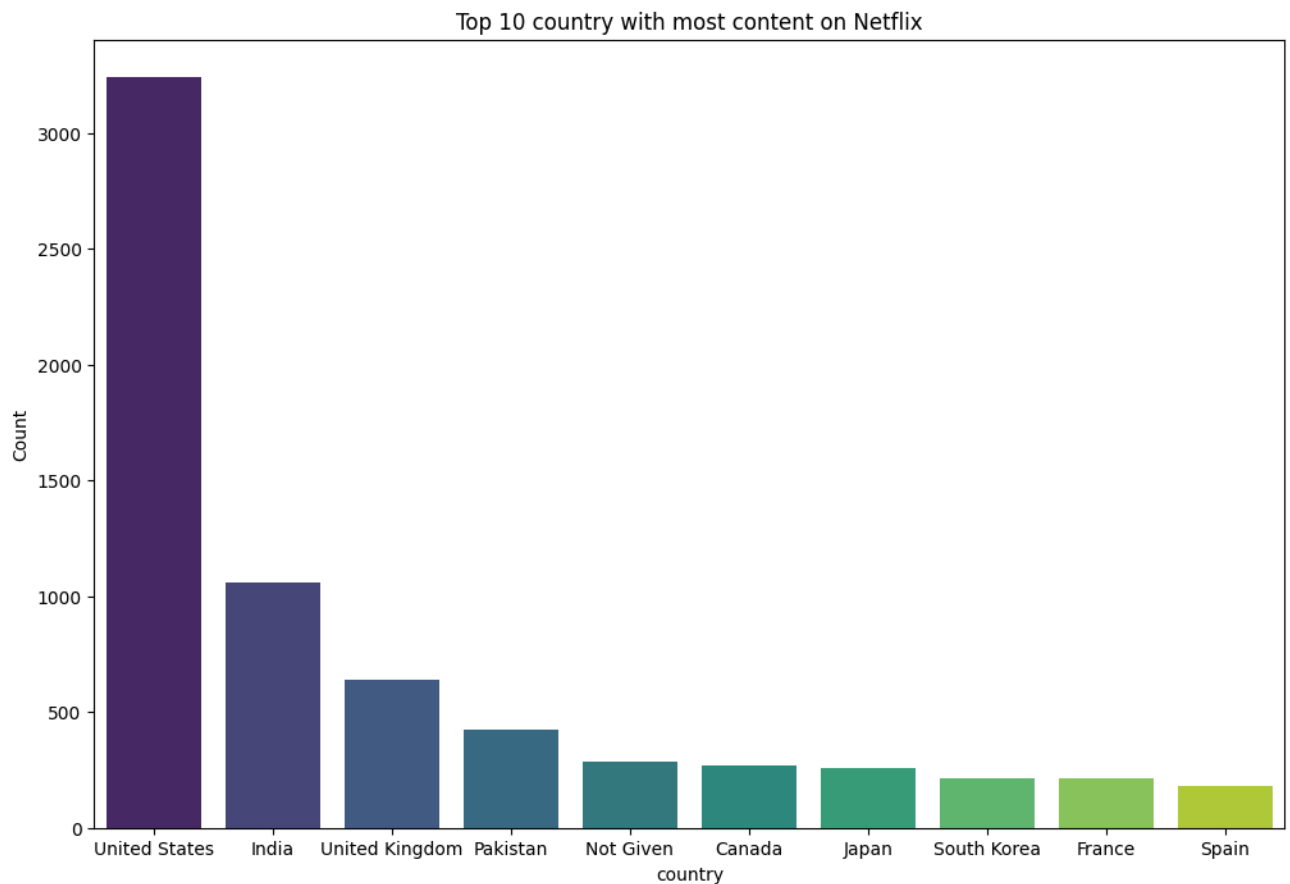
```
plt.figure(figsize=(12,8))
sns.barplot(x=top_10.index, y=top_10.values, palette="viridis")
plt.xlabel('country')
```

```
plt.ylabel("Count")
plt.title("Top 10 country with most content on Netflix")
plt.show()
```

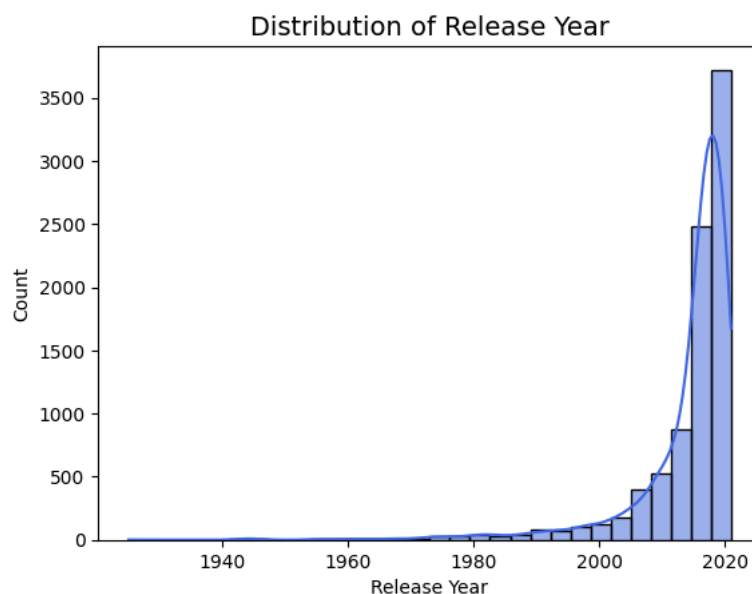
/tmp/ipython-input-3649572883.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

```
sns.barplot(x=top_10.index, y=top_10.values, palette="viridis")
```



```
sns.histplot(df['release_year'], bins=30, kde=True, color="royalblue")
plt.title("Distribution of Release Year", fontsize=14)
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.show()
```



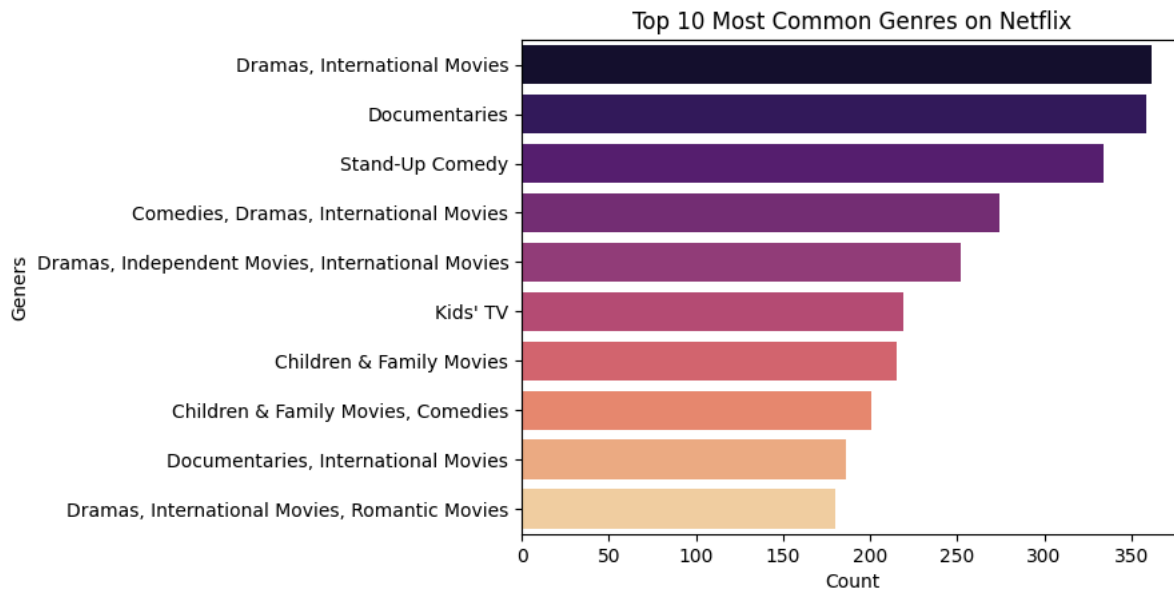
```
top_genres = df["listed_in"].value_counts().head(10)
sns.barplot(x=top_genres.index, y=top_genres.values, palette="magma")
```

```
plt.xlabel("Count")
plt.ylabel("Genres")
plt.title("Top 10 Most Common Genres on Netflix")
plt.show()
```

/tmp/ipython-input-2104863360.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set

```
sns.barplot(x=top_genres.values, y=top_genres.index, palette="magma")
```

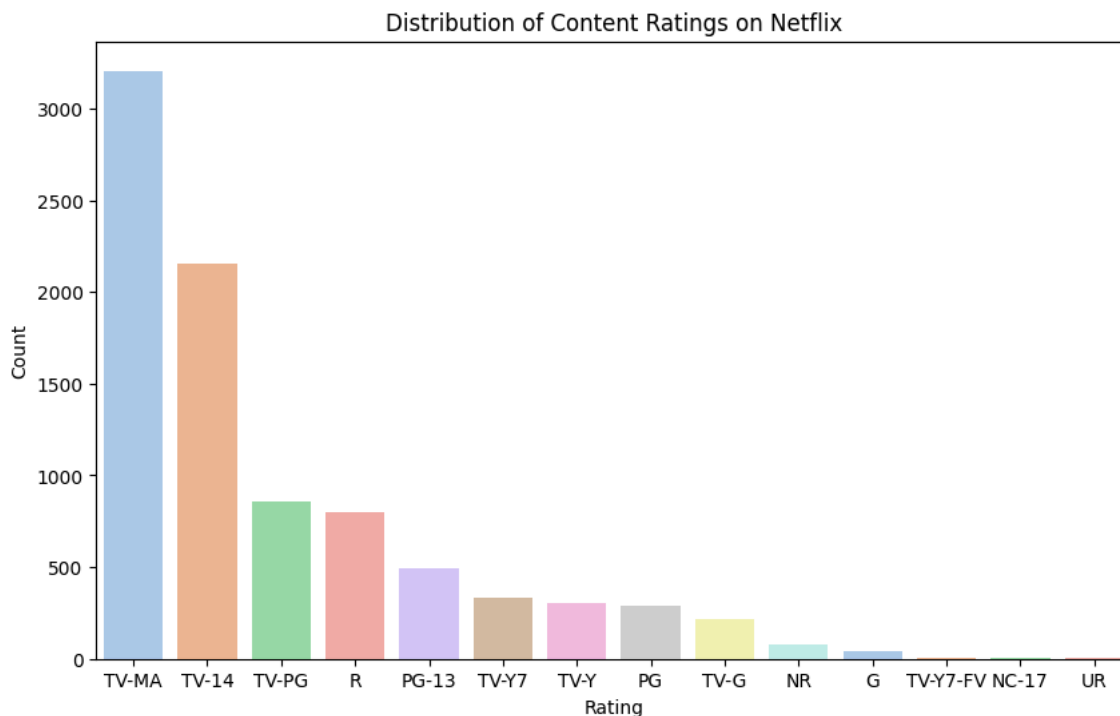


```
plt.figure(figsize=(10, 6))
rating_counts = df["rating"].value_counts()
sns.barplot(x=rating_counts.index, y=rating_counts.values, palette="pastel")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.title("Distribution of Content Ratings on Netflix")
plt.show()
```

/tmp/ipython-input-1892085084.py:3: FutureWarning:

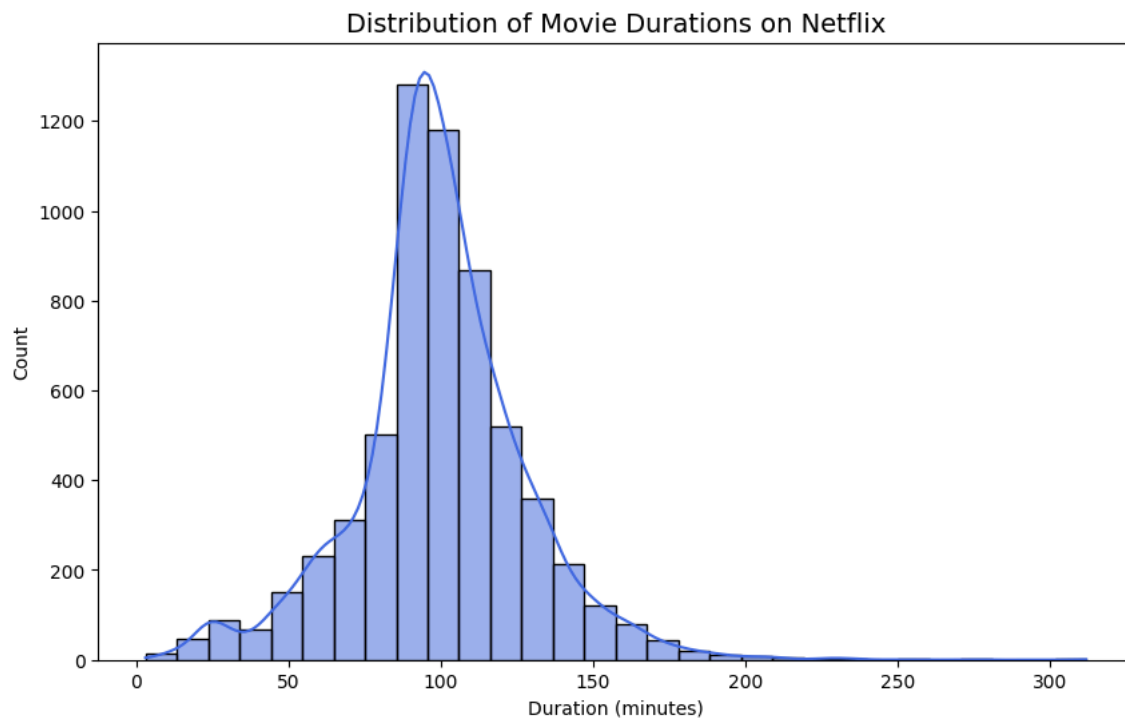
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

```
sns.barplot(x=rating_counts.index, y=rating_counts.values, palette="pastel")
```



```
movies_df = df[df["type"] == "Movie"].copy()
movies_df["duration"] = movies_df["duration"].str.replace(" min", "").astype(float)
```

```
plt.figure(figsize=(10, 6))
sns.histplot(movies_df["duration"], bins=30, kde=True, color="royalblue")
plt.title("Distribution of Movie Durations on Netflix", fontsize=14)
plt.xlabel("Duration (minutes)")
plt.ylabel("Count")
plt.show()
```

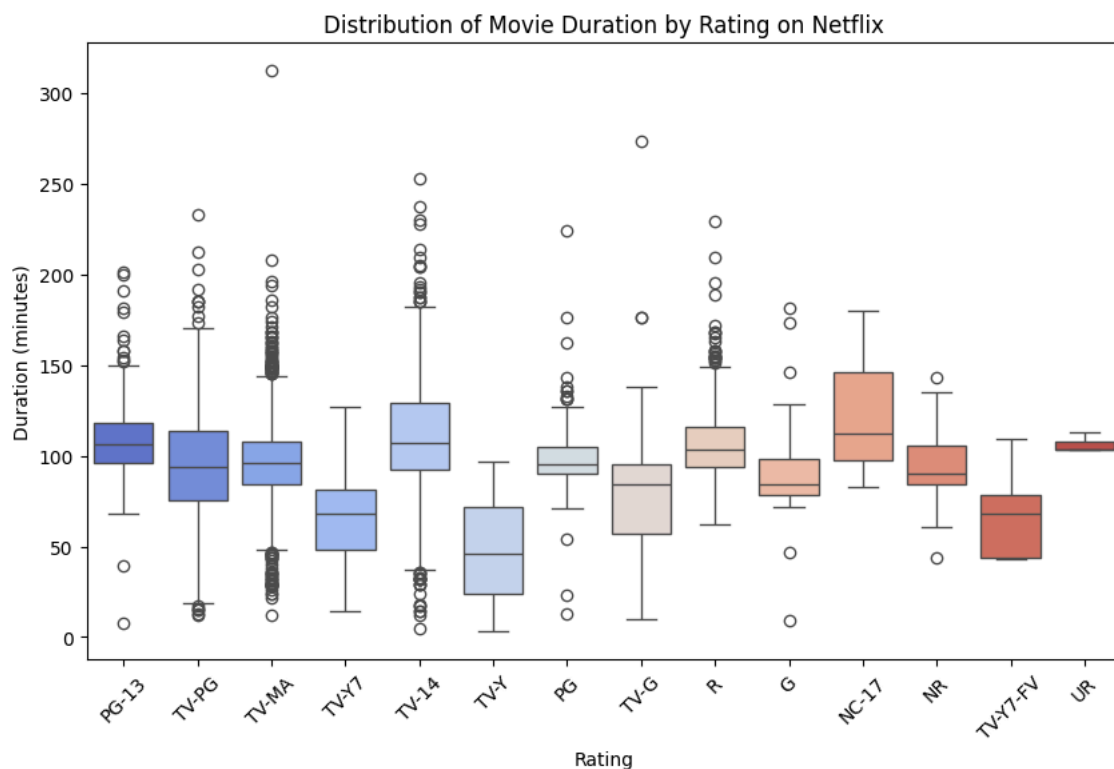


```
plt.figure(figsize=(10, 6))
sns.boxplot(data=movies_df, x="rating", y="duration", palette="coolwarm")
plt.xlabel("Rating")
plt.ylabel("Duration (minutes)")
plt.title("Distribution of Movie Duration by Rating on Netflix")
plt.xticks(rotation=45)
plt.show()
```

/tmp/ipython-input-482903113.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `hue\_order` to match the categories.

```
sns.boxplot(data=movies_df, x="rating", y="duration", palette="coolwarm")
```





```
df.columns
```

```
Index(['show_id', 'type', 'title', 'director', 'country', 'date_added',
      'release_year', 'rating', 'duration', 'listed_in'],
      dtype='object')
```

```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
a=['show_id', 'type', 'title', 'director', 'country', 'date_added',
   'release_year', 'rating', 'duration', 'listed_in']
for i in a:
    df[i]=le.fit_transform(df[i])
```

```
df.corr()
```

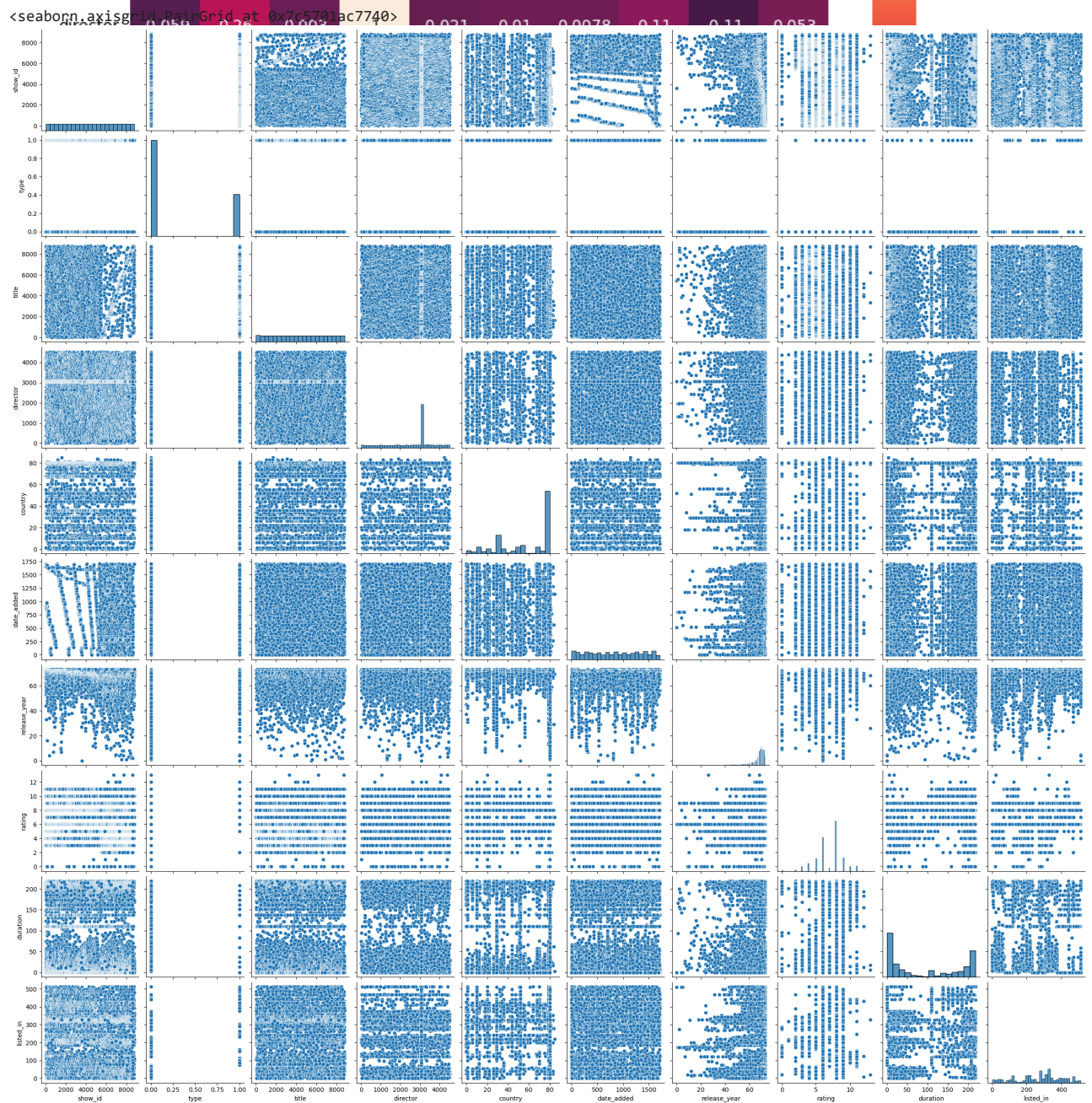
|              | show_id   | type      | title     | director  | country   | date_added | release_year | rating    | duration  | listed_in |
|--------------|-----------|-----------|-----------|-----------|-----------|------------|--------------|-----------|-----------|-----------|
| show_id      | 1.000000  | -0.127189 | 0.108164  | -0.059052 | 0.069123  | -0.008719  | -0.245564    | -0.127187 | 0.092445  | -0.072287 |
| type         | -0.127189 | 1.000000  | 0.026047  | 0.257602  | 0.042368  | 0.033827   | 0.185049     | 0.290778  | -0.381682 | 0.306382  |
| title        | 0.108164  | 0.026047  | 1.000000  | 0.003009  | 0.029428  | -0.002438  | -0.017079    | -0.017783 | -0.029480 | 0.016578  |
| director     | -0.059052 | 0.257602  | 0.003009  | 1.000000  | -0.021002 | 0.010269   | 0.007795     | 0.113582  | -0.113439 | 0.053042  |
| country      | 0.069123  | 0.042368  | 0.029428  | -0.021002 | 1.000000  | -0.036372  | -0.032319    | -0.116072 | 0.092700  | 0.044739  |
| date_added   | -0.008719 | 0.033827  | -0.002438 | 0.010269  | -0.036372 | 1.000000   | 0.078465     | 0.037562  | -0.005946 | -0.000128 |
| release_year | -0.245564 | 0.185049  | -0.017079 | 0.007795  | -0.032319 | 0.078465   | 1.000000     | 0.254172  | -0.005351 | 0.200234  |
| rating       | -0.127187 | 0.290778  | -0.017783 | 0.113582  | -0.116072 | 0.037562   | 0.254172     | 1.000000  | -0.021513 | 0.201639  |
| duration     | 0.092445  | -0.381682 | -0.029480 | -0.113439 | 0.092700  | -0.005946  | -0.005351    | -0.021513 | 1.000000  | -0.049587 |
| listed_in    | -0.072287 | 0.306382  | 0.016578  | 0.053042  | 0.044739  | -0.000128  | 0.200234     | 0.201639  | -0.049587 | 1.000000  |

## HEATMAP

```
plt.figure(figsize=(10,10))
sns.heatmap(df.corr(),annot=True)
plt.show()
```



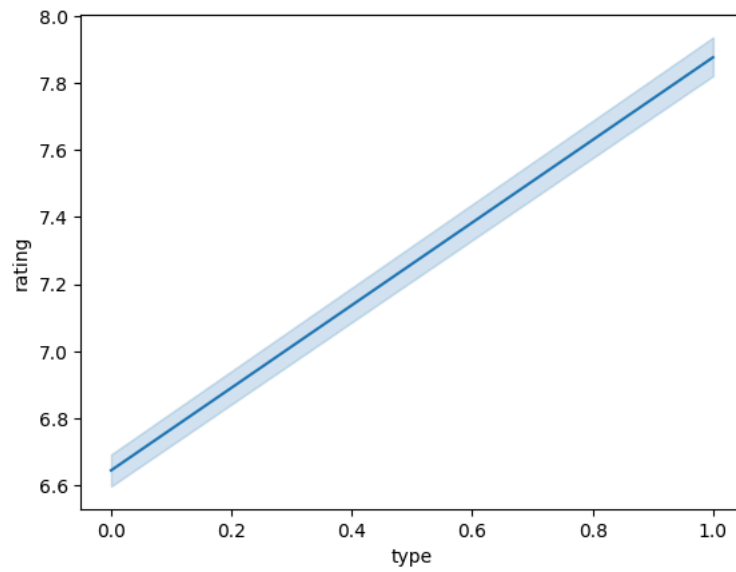
```
sns.pairplot(df)
```



Line Plot Of Type **vs** Rating

```
sns.lineplot(x='type',y='rating',data=df)
```

<Axes: xlabel='type', ylabel='rating'>



```
sns.lineplot(x='rating',y='duration',data=df)
```

<Axes: xlabel='rating', ylabel='duration'>

